often proprietary and even poorly documented. There are possibilities that without a common foundation for communication, we may be locked into just a single vendor for all of our devices, and worse, we may be left stranded with a pile of different non-functional hardware if the company which makes our devices goes defunct or decides to no longer support our devices.

There is no doubt that open source dominates large swaths of the intelligent networking and cloud platform software. And for that to translate into IoT dominance, developers will have to fill all possible gaps and implement technologies which are essential for IoT. Let's have a look at the role of open source in IoT and how it can help in handling the different challenges.

1. We are all aware that when we use different open source IoT frameworks, we spend no money since they are free for use. With costs not a barrier, everyone will implement IoT without any hesitation.
2. If we decide to adopt an open source IoT framework, we are not only drawing on the skills of the developers working on this but the whole open source community. Additionally, this wider support base often leads to developers getting inspired to build newer applications, which they might not have dreamt of when working in a closed, proprietary environment.
3. The success of the IoT market greatly lies in the connectivity of devices, which more often than not, share different hardware and operating systems. This is a serious obstacle to the very connectivity we are aiming for. If open source APIs are used for IoT, then we can get rid of this obstacle. This is because open source APIs offer a uniform gate for different software, hardware and the systems to communicate with one another.
4. If one chooses an open source IoT framework, then developers will be able to build different products, which will be interoperable across different OSs such as Android, Windows, iOS and Linux.
5. An open source IoT framework injects the software development life cycle with innovation and agility, which different proprietary models fail to match, since it offers a wide range of libraries, SDKs and open source hardware like Raspberry Pi and Arduino. With the help of an open source IoT framework, developers ensure that companies remain on the cutting-edge of technology by using different open sourced tools to customise IoT platforms to suit their needs.
6. If we look at the privacy challenge presented by IoT, open source software can protect individuals' data by implementing really strong encryption for the use of the general public (SSH, SSL, PGP, etc), and hence supply the building blocks for mobile security and the protection of data.
7. The history of open source software and security has been a roller coaster ride. After long years of debate, IT professionals finally began to appreciate the 'many eyes' approach of open source software communities, when it comes to detecting and addressing the security risks. The low defect rate of open source software code has been proved by independent studies such as Coverity Scans.
8. Huge amounts of data are produced by different hardware sensors integrated with IoT, and it becomes quite a challenge to handle this data and process it. This data does not just require a different scale of storage and processing, but it also requires new techniques like machine learning, artificial intelligence and data mining. These allow us to find patterns in the data that would not be obvious to different traditional analytics methods. Different open source Big Data tools make such analysis possible.

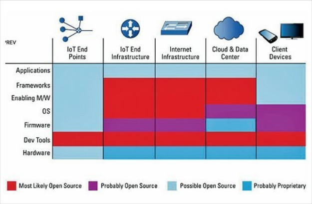

Figure 3: IoT open source - heat map (Image source: *googleimages.com*)

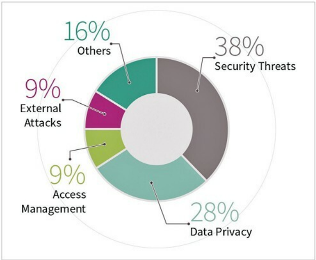

Figure 4: Threats to IoT (Image source: *googleimages.com*)

Security of IoT

According to a recent IOActive IoT security survey, less than 10 per cent of all IoT devices have adequate security. The most daunting threat to IoT is the ability of different hackers to